

Progress Test 2.A — Elec-S1

2.A · Characteristic curve and experimental evidence for Ohm's law

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Reading a slope

(1 mark)

The characteristic $i = f(u)$ of a resistor is a straight line through the origin, of slope 20 mA V^{-1} . Give its resistance R .

For $i = f(u)$: $i = \frac{u}{R}$, so the slope is $\frac{1}{R} = G = 20 \times 10^{-3} \text{ A V}^{-1}$
$R = \frac{1}{2.0 \times 10^{-2} \text{ A V}^{-1}}$ $R = 50 \text{ } \Omega$

Marking

①
0 1
☐ ☐

Marking scheme

① 1 pt $R = 50 \text{ } \Omega$. (Answering $R = 20$ or 0.02 confuses the slope with R .)

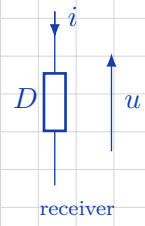
Examiner's tip

On $i = f(u)$ (u on the horizontal axis) the slope is $1/R = G$. On $u = g(i)$ it is R (Fig. 18 of the notes). Always check which quantity is on which axis.

Exercise 2. Characteristic of a component — definition

(5 marks)

Define the characteristic of a two-terminal component. Which convention is used to plot the characteristic of a passive component? Draw it and give the sign of the power received. What is the shape of the characteristic of an ohmic conductor?

The characteristic is the curve $i = f(u)$ giving the current i through the component as a function of the voltage u across it (u on the horizontal axis, i on the vertical axis). It is obtained point by point by measuring pairs (u_k, i_k) for various settings. Passive component: receiver convention. The current enters through the + terminal, so the arrows of u and i are opposite, and $p = ui > 0$ is <i>received</i>. Ohmic conductor: a straight line through the origin (slope $1/R$).	
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Marking

①
0 1 2
☐ ☐ ☐
②
0 1 2
☐ ☐ ☐
③
0 1
☐ ☐

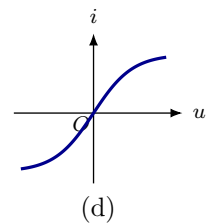
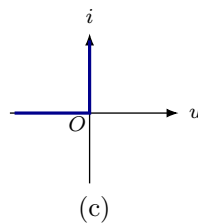
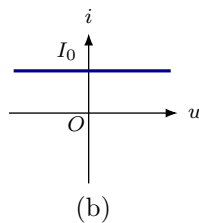
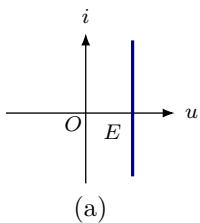
Marking scheme

- ① **2 pts** Curve $i = f(u)$ (1 pt), with the axes stated and measured point by point (1 pt).
- ② **2 pts** Receiver convention drawn with opposite arrows (1 pt); $p = ui > 0$ received (1 pt).
- ③ **1 pt** Straight line through the origin.

Exercise 3. Identifying characteristics

(3 marks)

The four characteristics below are plotted in the (u, i) plane.



Characteristic (a) is that of:

- | | |
|--|---|
| ☐ an ohmic conductor | ☒ an ideal voltage source |
| ☐ an ideal current source | ☐ a lamp |

Why: whatever the current, the voltage keeps the same value $u = E$: this is the definition of an ideal voltage source (vertical line). (b) is an ideal current source, (d) a lamp.

Characteristic (c) is that of:

- | | |
|--|--|
| ☐ a Zener diode | ☐ a lamp |
| ☒ an ideal diode | ☐ an ideal current source |

Why: no current for $u < 0$ (blocked), and $u = 0$ whatever $i > 0$ (conducting). A Zener diode would also conduct in reverse below $-U_Z$.

Which statement is correct?

- ☐ Every characteristic is a straight line through the origin.
- ☒ A straight-line characteristic through the origin is the signature of an ohmic conductor.
- ☐ The characteristic of a lamp is a straight line through the origin.
- ☐ A characteristic does not depend on how u and i are oriented.

Why: “Beware! Not all characteristics are straight lines through the origin” (notes, 2.A). The characteristic does depend on the convention chosen.

Exercise 4. Estimating a resistance from measurements

(5 marks)

The characteristic of a resistor was measured in the receiver convention. The measurements at low current are the noisiest.

i_k (mA)	1.0	2.0	4.0
u_k (V)	0.70	0.90	2.00

(a) Compute the naive estimate $R_{\text{naive}} = \frac{1}{N} \sum_k u_k / i_k$. (b) Compute the least-squares estimate $\hat{R} = \sum_k u_k i_k / \sum_k i_k^2$. (c) Which value should be kept? Justify.

(a) Ratios (V/mA = kΩ): $\frac{0.70 \text{ V}}{1.0 \text{ mA}} = 0.70 \text{ kΩ}$, $\frac{0.90}{2.0} = 0.45 \text{ kΩ}$, $\frac{2.00}{4.0} = 0.50 \text{ kΩ}$
$R_{\text{naive}} = \frac{0.70 + 0.45 + 0.50}{3} = \frac{1.65}{3}$ $R_{\text{naive}} = 0.55 \text{ kΩ} = 550 \text{ Ω}$
(b) $\sum u_k i_k = 0.70 \times 1.0 + 0.90 \times 2.0 + 2.00 \times 4.0 = 10.5 \text{ V mA}$
$\sum i_k^2 = 1.0^2 + 2.0^2 + 4.0^2 = 21 \text{ mA}^2$
$\hat{R} = \frac{10.5}{21} \text{ kΩ}$ $\hat{R} = 0.500 \text{ kΩ} = 500 \text{ Ω}$
(c) Keep $\hat{R} = 500 \text{ Ω}$. The relative uncertainty on u_k / i_k is dominated by that on i_k , which is largest at low current. The naive mean gives the noisy first point (0.70 kΩ) the same weight as the others; the regression weights each point by i_k^2 .

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1

Marking scheme

- ① 1 pt The three ratios, with a correct unit (V/mA = kΩ).
- ② 1 pt $R_{\text{naive}} = 0.55 \text{ kΩ} = 550 \text{ Ω}$.
- ③ 1 pt $\sum u_k i_k = 10.5 \text{ V mA}$ and $\sum i_k^2 = 21 \text{ mA}^2$.
- ④ 1 pt $\hat{R} = 500 \text{ Ω}$.
- ⑤ 1 pt $\hat{R}$ kept, justified (low-current points noisier / weighting by i_k^2).

Examiner's tip

V/mA gives kΩ, not Ω: a factor of 1000 is the most common slip here (Tab. 3 of the notes lists r_k in kΩ). Check: all points should lie close to $u = \hat{R} i$; here the residuals are +0.20, -0.10 and 0 V.

Your score: / 14 → mark out of 20 = score $\times$ 20/14 = / 20